



Data Analysis of NHS
Missed Appointments and Resource
Utilisation: Insights for Improved
Healthcare Efficiency

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August 2025

NHS Appointment Utilisation and Missed Appointments

Technical Report

The NHS faces increasing demand, constrained resources, and recurring missed appointments. With an ageing population, this puts strain on services. Missed appointments come at financial and social cost—each costing £30 and denying others care.

Data analysis was carried out on NHS appointment datasets and external sources to uncover patterns in utilisation, capacity, and missed appointments.

Python (with Matplotlib and Seaborn) was used to analyse data structure, trends, and insights. Twitter data was reviewed to understand public opinion and themes.

Objective

The NHS is asking:

1. Has there been adequate staff and network capacity?
2. What was the actual utilisation of resources?

#Analytical Approach

Four datasets were used:

- `actual_duration.csv`: Appointment durations, location, date, count
- `appointments_regional.csv`: Mode, date, location, time between booking and appointment, count
- `national_categories.xlsx`: Category of treatment, date, location, count
- `twitter.csv`: Tweets referencing NHS topics

The Excel file was converted to CSV for consistency.

All datasets were checked for missing values, duplicates, and cleaned accordingly.

Duplicates were found only in the `appointments_regional` dataset and removed.

Cleaned datasets were named with the `_clean` suffix.

Key findings:

- 106 locations across 7 regions
- General practice made up 91.5% of services

- Face-to-face was the most common mode, followed by telephone
- November 2021 saw the most appointments (30.4M); August 2021 the fewest (23.9M)
- Most appointments lasted around 10 minutes; 1–5 minutes were most frequent

Key Findings & Visualizations

1. Appointment Duration & Volumes

- The majority of appointments had a duration of 6-10 minutes. However, there are issues in recording this metric and indeed unknown/data quality makes up the largest count.
- Most appointments are same day

2. Service Setting & Context Type Trends

- GP appointments make up the bulk of appointments
- Face to face and telephone were the primary mode types, with telephone becoming more common during Covid.
- Appointment counts varied seasonally, with winter months (December to February) showing a dip, maybe due to holidays and weather impacts.

3. National Categories

- Most appointments were categorised as “Routine”.
- Seasonal fluctuations were more visible in urgent appointments, which tend to spike in winter.

4. Temporary/Unusual Trends

- The data for 2020 to 2022 showed :
 - A steady upward trend in total appointments, suggesting increasing demand.
 - A dip during Covid lockdowns, with gradual recovery afterward.
 - Evidence of infrastructure strain during peaks (e.g., winter and flu seasons).

5. Missed Appointments Analysis

- A small but significant percentage of appointments (4-5%) were missed across the period. There were a further 4-5% of appointments that were unmapped
- Missed appointments were more frequent in routine appointments than in urgent ones.
- These missed slots represent a significant cost burden and a lost opportunity to serve other patients.

6. Twitter Insights

- Top trending hashtags included #healthcare, #health and #medicine
- These insights could be used for sentiment analysis and to inform patient engagement strategies or targeted awareness campaigns. Indeed, patient views and concerns. However, the data as it stands has very limited use.

Capacity & Utilisation

- The NHS network is experiencing growth in demand, particularly in general practice settings.
- Capacity gaps are emerging during peak periods (e.g., winter), suggesting potential under-resourcing.
- A limitation found is that the 1.2 million capacity figure is used across all days, including weekends where appointment numbers are a lot lower. This then skews the data. Indeed, a similar situation arises with the count of Monday appointments due to bank holidays

Recommendations

1. Capacity Planning:

- Accurate and more reliable data is required. Unmapped or data quality are frequently the highest count in metrics
- Continue monitoring appointment volumes monthly and adjust staffing accordingly. The use of 1.2Million daily appointments needs to be amended based on demand and availability. Having this set for each day

including weekends makes no sense. Capacity should be available where it is needed, like a retailer assigns more staff to busier periods.

- Model future demand using historical trends, particularly focusing on winter spikes. However, the time period covered was unusual with the covid pandemic and likely wouldn't represent what is typical.
- Prioritise face-to-face and urgent care resourcing, where missed appointments are costlier.

2. Addressing Missed Appointments:

- The use of text or email reminders, especially for routine appointments may be beneficial. However, we need to remember that not all patients are tech savvy.
- Target high no-show regions or categories with public awareness campaigns informed by social media sentiment.
- Maybe make the process of making , rescheduling and cancelling appointments easier. Not everyone has the time or indeed wants to have to wait on the phone to speak to reception staff. Maybe the use of technology can play a part here through apps or patient portals. Where these are in place make sure they available for use

3. Social Media Integration:

- Social media has potential and is certainly something which should be looked into more. Not just Twitter but other social media apps.
- Develop dashboards to monitor **weekly sentiment trends** across major regions.
- Leverage positive campaigns (e.g., #ThankYouNHS) to counteract negative sentiment during crises.

4. Further Data Development:

- Collect demographic and socio-economic data alongside appointment records to better segment and understand at-risk groups. Who are the patients that do not attend appointments, the data doesn't tell us
- Collaborate with other data providers to incorporate transport, weather, or employment patterns, which may influence attendance.

Conclusion

The analysis indicates that while the NHS has managed to deliver increasing numbers of appointments, the network is under visible strain during peak periods. Missed appointments remain a persistent issue with substantial financial and service costs.

However, a big issue is that of unmapped and data quality. Without accurate, up to date and consistent data it's difficult to get a clear picture and this looks like the main issue holding the NHS back in terms of understanding capacity and utilisation.

Social media is underused for understanding and influencing public sentiment and behaviour.

The NHS may benefit from optimising existing capacity, looking at where demand is highest and utilising resources when required and scaling back when not needed.

The use of the same figure for capacity every day does not work and indeed skews the data into not giving a clear representation of the situation.

The use of tech, particular social media and rescheduling systems may help reduce the numbers of patients not attending.